

# Envelope detection based on online ICA algorithm and its application to motor imagery classification \*

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**Abstract**—Independent Component Analysis (ICA) is a promising tool for brain-computer interface (BCI). But most of ICA-based BCI researches only used batch ICA algorithm as the offline preprocessing step for EEG artifact removal and pattern enhancement. This paper explored a new approach of applying online ICA based on sliding window Infomax algorithm for BCI implementation. In addition to having good performance of blind source separation as traditional ICA, the proposed method has the characteristics that can synchronously realize the online envelope detection of multi-channel signals, which is that other methods do not have, such as Hilbert transform. Then the online ICA is applied to the envelope detection of mu rhythm evoked by motor imagery and good classification results of imagining left and right hand movement are achieved on real-life data.

## I. INTRODUCTION

Motor imagery EEG is a typical input signal of BCI system. Based on the detection and classification of mu rhythm amplitude changes caused by event-related (de)synchronization (ERD/ERS) during motor imagery, this form of BCI system can provide a non-muscular communication channel that enables people to control external computers or other electronic devices using their motor imagination[1][2]. So how to online extract the envelop changes of mu rhythm as the feature and classify the left and right hand movements correctly is critical for the real-time BCI system. Many signal processing methods in time-frequency domain (FFT, STFT, Wavelet, etc.) and the classifier such as RBF-NN, LDA and SVM have been used for the BCI system. However, those methods are mainly adopted to off-line analysis and not suitable for long-lasting recordings due to the strong non-stationary of EEG signal [3][4].

ICA-based BCI has become a hotspot in recent years, which is considered as a potential tool in BCI research [5]. However, many BCI researchers just adopted batch ICA

algorithm as EEG preprocessing method such as EOG, power-line removal and basic rhythm enhancement, but in BCI research, the application of online ICA algorithm is hardly reported. In this paper, the Infomax algorithm based on sliding window is employed to the envelope extraction of mu rhythm evoked by motor imagery, and the dynamic mixing matrix(DMM) can be used to classify the left and right hand movement effectively. This is a new thought of online ICA-based application in BCI and can also be extended to other signal processing fields.

## II. PROPOSED ALGORITHM

### A. Online Infomax algorithm based on sliding window

ICA is a kind of blind source separation method for multi-channel mixed signals. ICA assumes that a set of observations of random variables  $\mathbf{x}=[x_1(t), x_2(t), \dots, x_N(t)]^T$  are generated as a linear mixture of independent component  $\mathbf{x}=\mathbf{A}\mathbf{s}$ , where  $t$  is the time or sample index,  $\mathbf{A}$  is a mixing matrix and  $\mathbf{s}$  is the source signals. The aim of ICA is to find a linear transformation given by a separation matrix  $\mathbf{W}$  as in formulation  $\mathbf{u}=\mathbf{W}\mathbf{x}$ , so that the random variables  $u_i, i=1,2,\dots,N$  in the matrix  $\mathbf{u}$  are as independent as possible. In many different algorithm theories of ICA, extended-Infomax algorithm is a typical ICA algorithm based on the maximum information transmission theory and is widely used in blind source separation and blind deconvolution for multi-channel speech signals[6][7]. The learning rules of separation matrix  $\mathbf{W}$  respectively using batch Infomax algorithm and online Infomax algorithm updated with a single sample are given as (1) and (2):

$$\Delta \mathbf{W} = \mu \left\{ \mathbf{I} - E \left[ \mathbf{K} \tanh(\mathbf{u}) \mathbf{u}^T - \mathbf{u} \mathbf{u}^T \right] \right\} \mathbf{W} \quad (1)$$

$$\Delta \mathbf{W}_n = \mu \left[ \mathbf{I} - \mathbf{K}_n \tanh(\mathbf{u}_n) \mathbf{u}_n^T - \mathbf{u}_n \mathbf{u}_n^T \right] \mathbf{W}_n \quad (2)$$

Here, we propose the online Infomax algorithm by making use of sliding window approach (Fig.1). To all  $N$  channels, assuming the length of sliding window is  $L$ , when a new sample value  $x(n)$  (here  $n$  indicates the current time) is coming, the online learning rule is described by

$$\Delta \mathbf{W}_n = \mu \left\{ \mathbf{I} - \mathbf{K}_n \cdot E_L \left[ \tanh(\mathbf{u}_n) \mathbf{u}_n^T \right] - E_L \left[ \mathbf{u}_n \mathbf{u}_n^T \right] \right\} \mathbf{W}_n \quad (3)$$

where  $\mathbf{u}_n=[u(n-L+1), \dots, u(n-1), u(n)] \in R^{N \times L}$  indicates the  $L$  output values in the sliding window at time  $n$  according to  $\mathbf{u}=\mathbf{W}\mathbf{x}$  and  $E_L[\cdot]$  is to do the mean calculation for the  $L$ -wide data in the sliding window at time  $n$ . Here, using  $\mathbf{u}_n$  instead of the single value  $u(n)$  can help to improve the convergence performance of algorithm. Comparing (2) with (3), owing to be very sensitive to the instantaneous value of data for online

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algorithm updated with a single sample, the Infomax algorithm with sliding window has better stability and convergence performance.

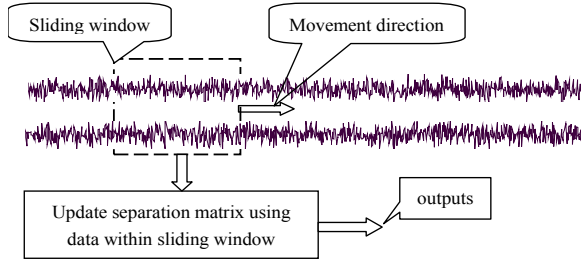


Figure 1. Description of online ICA based on Infomax algorithm with sliding window

Comparing with the traditional approach, the improved online Infomax algorithm has several important features as follows: first of all, the online method can perform the envelope detection based on dynamic mixing matrix(DMM) coefficients  $\mathbf{A}(n)=\mathbf{W}^{-1}(n)=[a_{ij}(n)] \in R^{N \times T}, 0 \leq n \leq T-1, 1 \leq i, j \leq N$ , where  $T$  is the total sample length. This important feature can be used in many envelope detection problems. Secondly, besides having the inherent advantages of ICA, such as blind source separation, the proposed algorithm can also deal with the time-variant mixing ICA model which is often hard to do well using the traditional ICA algorithm.

### B. Simulation example of envelope detection

There are two amplitude modulation signals as in Fig.2(a),

$$\begin{aligned} s_1(t) &= c_1(t) \sin(20\pi t) \\ s_2(t) &= c_2(t) \sin(30\pi t) \end{aligned} \quad (4)$$

where  $c_1(t)$  and  $c_2(t)$  are a sinusoidal signal and a square wave respectively with low frequency. The length of signal equals 3000 samples and sampling rate is 128Hz. The two AM signals are mixed by a random  $2 \times 2$  matrix  $\mathbf{B}=[b_{11}, b_{12}; b_{21}, b_{22}]$ (seen as Fig.2(b)). Comparing the separation results in Fig.2(c) with results in Fig.2(e), it can be seen that online ICA algorithm based on sliding window can dynamically separate the carrier signals with amplitude equalized from the linear mixed signals, but the traditional batch algorithm can only separate the source modulated signals. In Fig.2(d), it shows the change of the DMM coefficients  $\{a_{11}(t), a_{12}(t), a_{21}(t), a_{22}(t)\}$ , which can express clearly the envelope change of the two modulated signals according to (5):

$$\begin{aligned} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} &= \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} s_1(t) \\ s_2(t) \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} c_1(t)\hat{s}_1(t) \\ c_2(t)\hat{s}_2(t) \end{bmatrix} \\ &= \begin{bmatrix} b_{11}c_1(t) & b_{12}c_2(t) \\ b_{21}c_1(t) & b_{22}c_2(t) \end{bmatrix} \begin{bmatrix} \hat{s}_1(t) \\ \hat{s}_2(t) \end{bmatrix} = \begin{bmatrix} a_{11}(t) & a_{12}(t) \\ a_{21}(t) & a_{22}(t) \end{bmatrix} \begin{bmatrix} \hat{s}_1(t) \\ \hat{s}_2(t) \end{bmatrix} \end{aligned} \quad (5)$$

Here, if mixing matrix  $\mathbf{B}$  is the identity matrix, the DMM coefficients  $\{a_{11}(t), a_{22}(t)\}$  are corresponding to  $c_1(t)$  and  $c_2(t)$ . These results indicate the proposed online ICA algorithm not only can realize the blind source separation, but also can real-time detect the envelope information by the DMM coefficients.

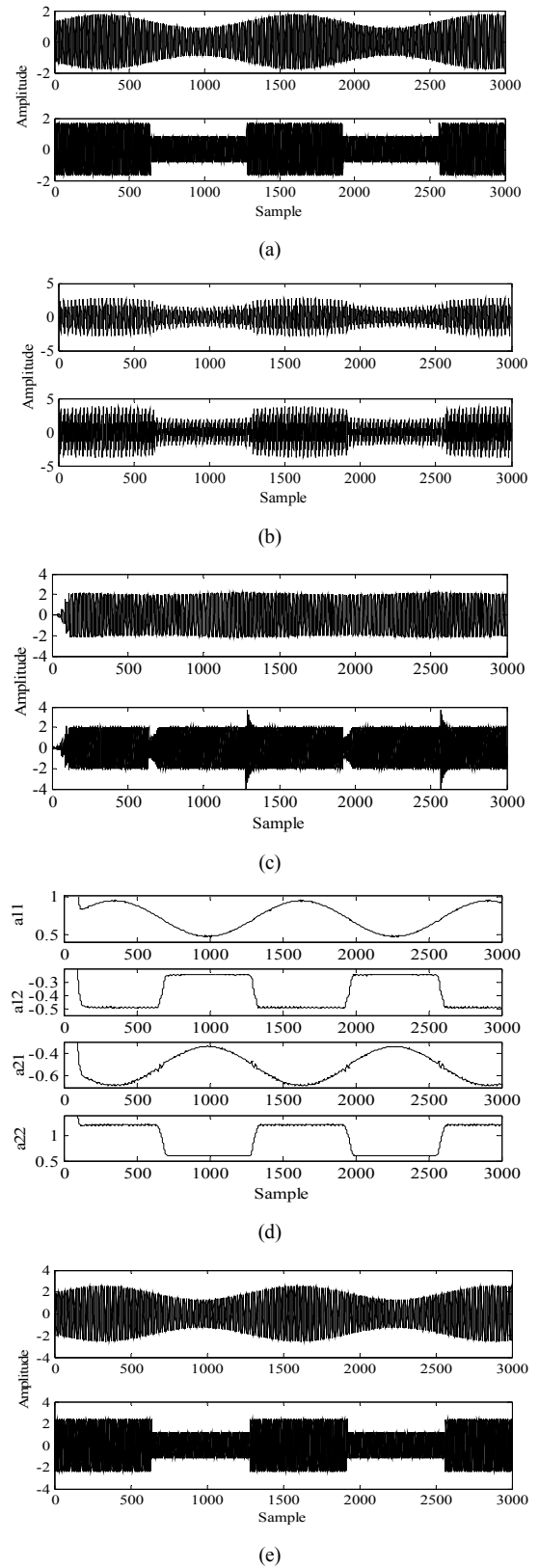


Figure 2. Simulation results for amplitude modulation signals: (a) two source signals with modulation; (b) the mixed signals by a random  $2 \times 2$  matrix; (c) the dynamic outputs based on the sliding window ICA method; (d) the dynamic mixing matrix coefficients, the curves from top to bottom are  $a_{11}(t)$ ,  $a_{12}(t)$ ,  $a_{21}(t)$ ,  $a_{22}(t)$ . (e) the output results of batch ICA algorithm.

### III. APPLICATION ON MOTOR IMAGERY CLASSIFICATION

#### A. Analysis of BCI competition data

The raw motor imagery EEG data were provided by Graz University of Technology in BCI competition IV data sets 2b. The data set consists of EEG data from 9 normal subjects. For each subject 5 sessions are provided, in which the last two sessions recorded with feedback were used in this paper. Each trial lasted about 9 seconds as shown in Fig.3. The imagery period is between 3s and 7.5s. The detailed experimental description can be available on open internet [8]. The data were sampled with 250Hz through three bipolar channels (C3, Cz, C4), bandpass-filtered between 0.5 and 100Hz and notch-filtered at 50 Hz.

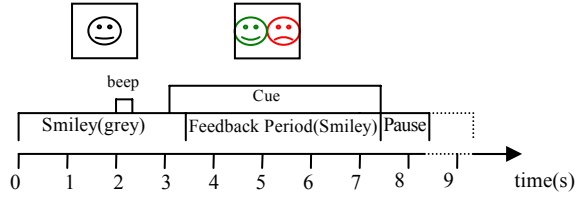


Figure 3. Timing scheme of one trial with smiley feedback

To a practical BCI system, it can be hoped to use less data from the minimum channels so as to obtain the classification results as better as possible. So in this paper, the data from two channels (C3, C4) are only used for analysis. The specific process for motor imagery classification can be shown in Fig.4.

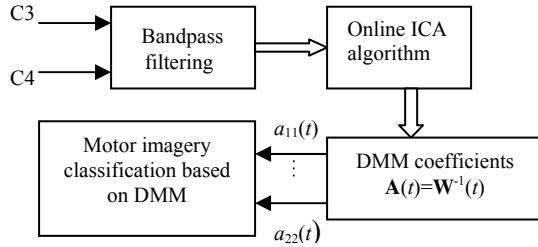
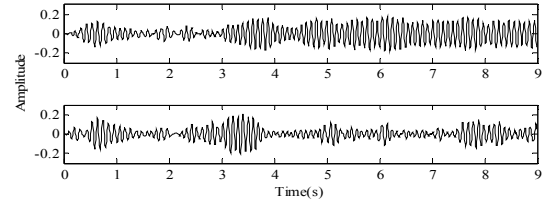


Figure 4. Schematic for motor imagery classification based on sliding window ICA algorithm

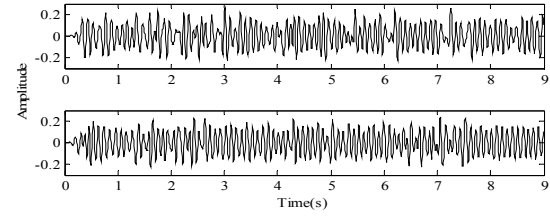
Here, the data of one trial by imagining left hand movement are randomly selected. In Fig.5(a), it shows the filtered EEG data between 8 and 14Hz on electrode C3 and C4. Then the online ICA algorithm is applied to the filtered data. In Fig.5(b), the dynamic outputs have the feature of so-called “amplitude equalization”, and the square value of DMM coefficients is shown in Fig.5(c), which can reflect the energy changes of mu rhythm’s envelope. So by comparing the sum of energy value obtained from DMM coefficients on C3 with that on C4, we can realize the classification of left and right hand motor imagery tasks. Because the imagery task of each trial starts at 3s, here we choose the square sum within certain time (3.5s-7s) of DMM coefficients as features for classification in (6), and the classification rule is defined as in (7):

$$E = [e_{C3}, e_{C4}]^T = \left[ \sum_t \{a_{11}^2(t) + a_{12}^2(t)\}, \sum_t \{a_{21}^2(t) + a_{22}^2(t)\} \right]^T \quad (6)$$

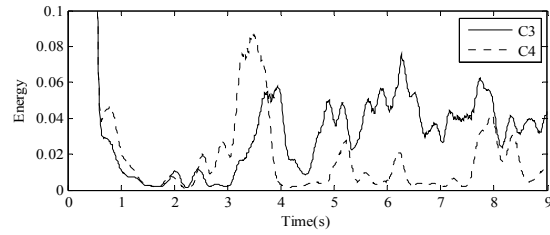
$$e_{C3} - e_{C4} \begin{cases} > 0 & \text{motor imagery of left hand} \\ < 0 & \text{motor imagery of right hand} \end{cases} \quad (7)$$



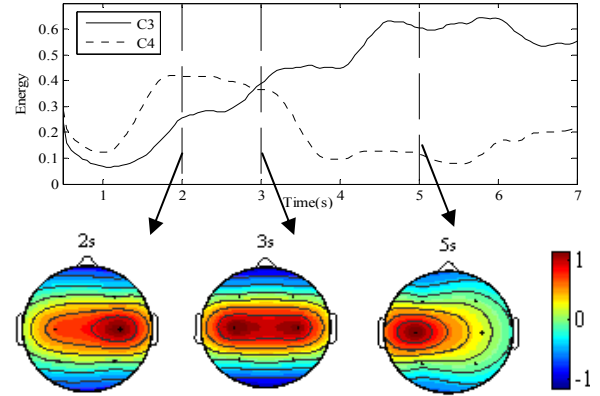
(a)



(b)



(c)



(d)

Figure 5. The application of proposed algorithm on real-world motor imagery EEG: (a) one trial filtered data( 8-14Hz) of left hand motor imagery EEG on electrode C3 and C4; (b) two dynamic separation signals with amplitude equalized; (c) The square value of DMM coefficients on C3 and C4; (d) The brain topographic map corresponding different time

For further studying the dynamic distribution of mu rhythm on the electrodes C3 and C4 in process of motor imagination, the moving average method by a 2s window is used to the square value of DMM coefficients, and also the brain topographic map corresponding different time (2s, 3s and 5s) are given in Fig.5(d). From the results, it can be seen that the ERD/ERS phenomena obviously appears after 3s during the left hand motor imagery.

Table 1 gives the classification accuracy based on the proposed online ICA algorithm for BCI competition IV data sets 2b. In order to test the validity of the proposed method, the Hilbert transform, which is known as a typical method for envelope detection, and the energy method based on second order moment are respectively used to the BCI data. From the classification results, it can be seen that the proposed online ICA algorithm is more effective than the other two methods besides “B0605E” data. It must be mentioned here that this classification approach do not need using the training data to train the classifier with LDA, SVM etc, but just need utilizing the DMM coefficients as envelope feature of mu rhythm to classify. In addition, only two channels of C3 and C4 are used for two kinds of classification in the present study so as to improve the practicability and portability of the BCI system. Certainly, the classifier and the optimal number of channels should be further considered for the recognition of multi-class mental tasks.

TABLE I. CLASSIFICATION RESULTS FOR BCI IV DATA SET 2B BY DIFFERENT ALGORITHMS (ACCURACY (%))

| data | algorithm |                      |                                   |                   |
|------|-----------|----------------------|-----------------------------------|-------------------|
|      |           | Online ICA algorithm | Online Second order Moment Energy | Hilbert Transform |
| S1   | B0104E    | 78.75                | 76.88                             | 76.88             |
|      | B0105E    | 66.25                | 65                                | 65.63             |
| S2   | B0204E    | 60                   | 55.63                             | 52.50             |
|      | B0205E    | 60.62                | 56.87                             | 54.37             |
| S3   | B0304E    | 58.75                | 54.37                             | 51.88             |
|      | B0305E    | 58.75                | 56.87                             | 56.25             |
| S4   | B0404E    | 97.5                 | 96.88                             | 96.88             |
|      | B0405E    | 91.25                | 87.50                             | 90                |
| S5   | B0504E    | 83.75                | 83.13                             | 82.50             |
|      | B0505E    | 66.87                | 65                                | 64.38             |
| S6   | B0604E    | 80                   | 78.75                             | 78.13             |
|      | B0605E    | 78.13                | 79.37                             | 78.75             |
| S7   | B0704E    | 73.12                | 70.63                             | 73.12             |
|      | B0705E    | 77.5                 | 76.25                             | 75.62             |
| S8   | B0804E    | 71.25                | 70                                | 68.13             |
|      | B0805E    | 91.87                | 90                                | 84.38             |
| S9   | B0904E    | 85.62                | 82.5                              | 85                |
|      | B0905E    | 86.25                | 84.38                             | 84.38             |

### B. Analysis of experimental data

For further research of the physiological significance of motor imagery, we recorded the experimental data by ourselves in our own laboratory. The process design of one trial is similar to the Fig.3. During the recordings, each trial started with a fixation cross presented at the center of the monitor and an additional short acoustic stimulus at 2s. At 3s an arrow (left or right) was displayed for 1s as a cue. At the same time the subject was asked to imagine a corresponding hand movement for moving a bar into the direction of the cue. Each trial lasted 9 seconds.

The experiment data are selected from two subjects (S1, S2), and the data were recorded under three modes with different motor imagery tasks, which are respectively all left hand imagery, all right hand imagery and random combination with left and right hand imagery. Each group of data from different task consists of 100 trials and it was sampled with 250Hz through three bipolar channels (C3, Cz, C4) and bandpass-filtered between 0.5 and 70Hz. From the results in

Table 2, it can be shown that the classification accuracy rate of all left/right hand imagination tasks is generally higher than that of mixed tasks. By analysis of the potential reason, it is possible that the conversion process between two kinds of tasks does not exist during the imagination of all left/right tasks. And the lower accuracy rate of mixed tasks maybe caused by the conversion of mental tasks. To this point, it will be further researched in another paper.

TABLE II. CLASSIFICATION RESULTS FOR DATA RECORDED BY OURSELVES BASED ON ONLINE ICA ALGORITHMS (ACCURACY (%))

| tasks\data | S1    |       | S2    |       |
|------------|-------|-------|-------|-------|
|            | Data1 | Data2 | Data1 | Data2 |
| All left   | 98    | 95    | 99    | 87.5  |
| All right  | 90    | 97    | 91    | 86    |
| mixed      | 80.33 | 87    | 64    | 70    |

## IV. CONCLUSION

In this paper, the online envelope detection based on the sliding window ICA algorithm is presented and used to motor imagery EEG classification. Comparing with Hilbert transform method, the online ICA algorithm can synchronously process multi-channel signals, but Hilbert transform just is a batch process algorithm for envelope detection. Moreover, the proposed algorithm not only maintains all the advantages of the traditional ICA algorithm, but also can self-adaptively separate the blind source signals, which is a merit that other envelope detection methods do not have. Based on these characteristics, the proposed method is applied to the amplitude detection of mu rhythm evoked by motor imagery and the classification of left and right hand movement based on DMM. By extensive experiments on real-world data, the proposed algorithm is proved to be more feasible and practical than traditional ICA algorithms for BCI implementation with the validity and good performance.

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